

AUTOMATION TEST PLAN

Planning, scope, Jira workflow and execution strategy

JIRA-READY | SELENIUM | TESTNG | MAVEN | JENKINS | DOCKER

Prepared By	Pratha Khare
Project	ParaBank Automation Testing Framework
Document ID	PB-TP-PRK-001
Version	1.0
Date	June 2026

Document Control

Version	Date	Prepared By	Change Description
0	June 2026	Pratha Khare	Initial Jira-ready issue and testing document

Document reference: PB-TP-PRK-001. This document is prepared from the supplied ParaBank Eclipse project, its TestNG suite and the latest available execution artifacts.

1. Purpose

This test plan defines the approach for validating the ParaBank banking application through the Java Selenium automation framework prepared by Pratha Khare. It covers scope, test levels, test data, environments, execution, reporting, Jira traceability, risks, responsibilities and release criteria.

2. Test Objectives

- Validate that a new customer can register with a unique username.
- Confirm that stored credentials can be used for successful authentication.
- Verify that a logged-in customer can open a new account.
- Validate account overview visibility and core account information.
- Confirm that funds can be transferred between eligible accounts.
- Validate bill payment using Excel-driven payee and account data.
- Generate repeatable evidence through TestNG, Extent Reports, logs and screenshots.
- Create traceability between test scenarios, executions and defects in Jira.

3. Scope

3.1 In Scope

- Registration
- Login
- Open New Account
- Account Overview
- Transfer Funds
- Bill Payment
- Excel data read/write
- Screenshot and report generation
- Maven, Jenkins and Docker execution verification

3.2 Out of Scope

- Production banking integrations
- Real payment gateway processing
- Database-level validation
- Performance, penetration and accessibility certification
- Mobile native application testing
- Third-party email or SMS verification

4. Test Items and Jira Mapping

Jira Story	Feature	Automated Test	Coverage	Priority
------------	---------	----------------	----------	----------

Jira Story	Feature	Automated Test	Coverage	Priority
PB-STORY-01	Customer Registration	RegistrationTest.registerUser	Automated	High
PB-STORY-02	Customer Login	LoginTest.loginTest	Automated	Highest
PB-STORY-03	Open New Account	OpenAccountTest.createAccount	Automated	High
PB-STORY-04	Account Overview	AccountOverviewTest.accountOverviewTest	Automated	High
PB-STORY-05	Transfer Funds	TransferFundsTest.transferFundsTest	Automated	Highest
PB-STORY-06	Bill Payment	BillPayTest.billPayTest	Automated	Highest

5. Test Approach

The framework uses the Page Object Model. Test classes call reusable page actions, while utilities handle configuration, Excel operations, waits, screenshots and reports. TestNG controls suite execution and assertions. Maven provides repeatable command-line execution. Jenkins and Docker support continuous and isolated execution.

Functional testing: Validate expected behaviour of each banking module.

Data-driven testing: Read registration, login, transfer and bill-pay data from Excel.

Regression testing: Run the full six-test suite after framework or locator changes.

Negative testing: Recommended expansion for invalid login, invalid amount and mandatory-field validations.

CI execution: Execute through Maven in Jenkins and archive reports and screenshots.

6. Test Environment

Component	Planned Configuration
Operating System	Windows 11 for local execution; Linux container for Docker
Browser	Google Chrome
Language	Java
Automation	Selenium WebDriver
Test Framework	TestNG
Build Tool	Maven
Test Data	ParaBankData1.xlsx via Apache POI
Reporting	Extent Reports and TestNG/Surefire reports
Logging	Log4j2
CI/CD	Jenkins Pipeline
Issue Tracking	Jira

7. Test Data Strategy

- Registration username is generated dynamically to avoid duplicate-user conflicts.
- Generated username and password are written to the LoginData sheet for later modules.
- Transfer amount is read from the TransferData sheet.
- Bill-pay details are read from the BillPayData sheet.
- Sensitive values must not be printed to console or committed in public repositories.
- Test data should be reset or regenerated before each full regression cycle.

8. Entry and Exit Criteria

Entry Criteria	Exit Criteria
Build compiles successfully	All planned automated tests executed
ParaBank URL is reachable	No unresolved Highest/Critical defects
Chrome and driver are available	All failures analysed and linked to Jira
Excel sheets contain required data	Reports, logs and screenshots archived
Jira stories and acceptance criteria are available	Test summary reviewed and approved

9. Defect Management in Jira

Each failed scenario must be analysed before creating a defect. Application defects, framework defects and environment incidents must be classified separately.

Field	Required Jira Content
Sue Type	Bug
Summary	Module + concise failure statement
Description	Business impact and observed behaviour
Environment	URL, OS, browser, Java, Selenium, execution mode
Steps to Reproduce	Numbered and repeatable
Expected Result	Acceptance-criteria behaviour
Actual Result	Observed behaviour
Priority / Severity	Based on business and technical impact
Evidence	Screenshot, console log, Extent Report, video if required
Links	Story, test case, build and related defects

10. Jira Workflow

Recommended workflow: Backlog -> To Do -> In Progress -> Ready for QA -> Retest -> Done. Defects that cannot be reproduced should move to Needs Information rather than being closed without evidence.

11. Risks and Mitigation

Risk	Impact	Mitigation
Public ParaBank instability	False failures or internal server errors	Recheck manually, capture evidence and rerun after a controlled wait
Dependent test data	Later modules fail after registration/login failure	Preserve suite order and improve test-data setup
Fixed Thread.sleep calls	Slow or flaky tests	Replace with explicit waits
Weak assertions	False positive results	Add business assertions to every test
Browser/version mismatch	Session creation or CDP warnings	Keep Selenium and Chrome compatible
Sensitive data in logs	Credential exposure	Mask or remove password output

12. Deliverables

- Automation Test Plan
- Java Selenium source code
- TestNG XML suite
- Excel test data
- Extent and Surefire reports
- Screenshots and logs
- Test Analysis Report
- Jira Bug Report / Defect Register

13. Approval

Role	Name	Status	Date
Prepared By	Pratha Khare	Prepared	June 2026
Trainer / Reviewer	_____	Pending	_____
Project Owner	_____	Pending	_____